

Optimizing Inventory Management

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Abstract

This report presents a data-driven approach to optimizing inventory management for Amazon laptops in the Brantford region. The analysis incorporates demand forecasting, inventory control models, and optimization techniques to reduce costs, enhance stock availability, and improve decision-making processes.

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1. Introduction

Managing inventory efficiently is essential for retail businesses as it directly impacts customer satisfaction, operational efficiency, and overall profits. For a retail chain, maintaining the right balance of inventory ensures that products are available when customers need them while avoiding excess stock and unnecessary costs. This project focuses on improving inventory management for laptops at the Brantford location.

The main goal is to analyze sales data, inventory records, and customer demand to develop a system that maintains optimal stock levels. By doing so, we aim to lower costs and improve the availability of laptops. To achieve this, we will use data analysis, forecasting, and optimization techniques to create effective inventory strategies.

This report explains the approach, analysis, and results of the study. It provides practical recommendations to help the Brantford location reduce stock shortages, avoid overstocking, and improve overall inventory management for laptops.

2. Methodology

The methodology for this project involves systematically collecting, analyzing, and optimizing data to enhance inventory management for laptops at the Brantford location. The steps include data collection, exploratory analysis, forecasting, and optimization.

2.1 Data Collection

Data for this project was collected from the retail chain’s database, which includes information on laptop sales, inventory levels, and related product details.

The dataset contains the following columns -

Column	Description
ASIN	Unique Identifiers for each product
URL	Link to the product webpage for reference
Title	Name or description of the product
Memory	Memory Capacity of the laptop (e.g., 8GB, 16 GB).
Storage	Storage Capacity of the laptop (e.g., 256GB, 512 GB).
Brand	Manufacturer of the laptop (e.g., Apple, HP, Dell).
Star	Average customer rating out of 5
Rating_count	Total number of customer reviews for the product
Price	Cost of the laptop in CAD
Variant_attribute	Product variation
Purchase_August, Purchase_September, Purchase_October	Monthly sales data for August, September, and October
In_stock	Number of units currently available in inventory
Ship	Shipping method or availability
Seller	Vendor or seller offering the product

2.2 Data Cleaning and Preprocessing

The raw data was cleaned to address missing values, inconsistencies, and duplicate entries. Numerical values were standardized, and categorical variables were encoded to prepare the data for analysis. This step ensured the dataset was accurate and ready for use in subsequent modelling.

2.3 Exploratory Data Analysis (EDA)

EDA was performed to uncover patterns and trends in the data. Key metrics such as average sales, demand fluctuations, and inventory turnover rates were analyzed. Visualizations like histograms, scatter plots, and line graphs were used to highlight relationships between variables such as price, rating, and sales performance.

2.4 Forecasting Demand

Demand forecasting was conducted using statistical and machine learning models to predict future sales. Techniques such as time series analysis and regression modelling accounted for seasonality, sales trends, and external factors. These forecasts provided a foundation for inventory planning.

2.5 Optimization Modelling

Optimization models, including linear programming (e.g., Linear Regression) and non-linear algorithms (e.g., Decision Tree, Random Forest), were developed to determine the optimal inventory levels. The models aimed to minimize costs associated with overstocking and stockouts while ensuring sufficient stock to meet customer demand. Constraints such as supplier lead times, storage capacity, and reorder thresholds were incorporated into the models.

3. Data Analysis and Findings

3.1 Data Analysis

The analysis of the dataset provided valuable insights into sales trends, inventory patterns, and factors influencing customer demand for laptops. Below are the key steps and results of the data analysis:

3.1.1 Sales Trends

Monthly sales data for August, September, and October revealed fluctuations in demand. Sales in August were stable, while September and October saw increased demand, possibly due to seasonal promotions and back-to-school shopping.

3.1.2 Inventory Patterns

The `In_stock` column showed that inventory levels varied significantly across various products. Certain high-demand laptops had frequent stockouts, while slower-moving items had excessive stock levels, leading to higher holding costs.

3.1.3 Product Attributes and Sales

- **Price:** Sales were inversely related to price, with lower-priced laptops showing higher purchase volumes.
- **Brand:** Popular brands like Apple and Dell had higher sales compared to niche brands, indicating customer preference for well-known manufacturers.
- **Stars and Rating_count:** Products with higher ratings and more reviews showed better sales performance, highlighting the importance of customer feedback.

3.1.4 Variant Analysis

The `Variant_attribute` column revealed that specific configurations (e.g., higher memory and storage) were more popular, indicating a preference for mid-range to high-performance laptops.

3.1.5 Supplier and Shipping Analysis

Analysis of the `Ship` and `Seller` columns showed that faster shipping methods and trusted sellers were associated with higher sales. This indicates that delivery speed and seller reputation significantly influence purchase decisions.

3.2 Findings

3.2.1 Seasonal Demand Patterns

Sales showed clear seasonal patterns, with a notable increase in September and October. This suggests that inventory planning should consider seasonal trends to prevent stockouts during high-demand periods.

3.2.2 Stock Imbalances

Certain products were consistently out of stock, while others had excess inventory. This imbalance points to the need for better demand forecasting and inventory optimization to improve stock turnover.

3.2.3 Customer Preferences

Customers showed a preference for laptops from well-known brands with mid-range specifications, high ratings, and positive reviews. Pricing strategies and stock prioritization should align with these preferences.

3.2.4 Impact of Delivery and Seller Reputation

Products sold by reputable sellers with fast shipping options performed better. Improving partnerships with trusted suppliers and ensuring prompt delivery can boost sales and customer satisfaction.

4. Data Visualization

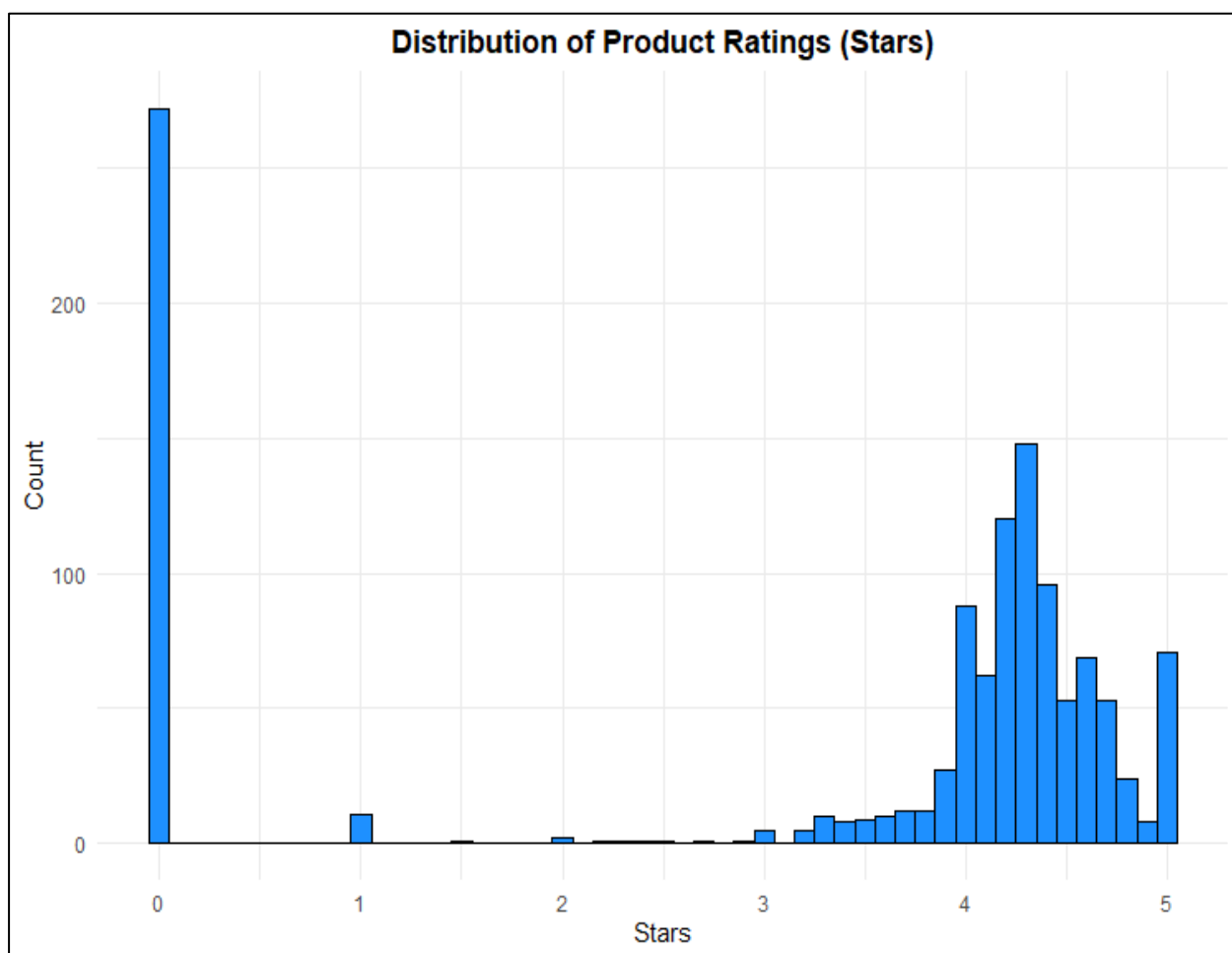
Data visualization was performed to extract actionable insights from the dataset and communicate trends, correlations, and patterns effectively. Various charts and graphs were created using **R** and libraries like **ggplot2**, ensuring clarity and interpretability.

4.1 Key Visualizations and Interpretations

4.1.1 Distribution of Product Ratings

- **Visualization:** A histogram of the Stars column revealed the distribution of product ratings.
- **Findings:** Most products had ratings clustered between 3.5 and 4.5 with few products having a 0 rating, indicating general customer satisfaction with room for improvement in product quality or experience.

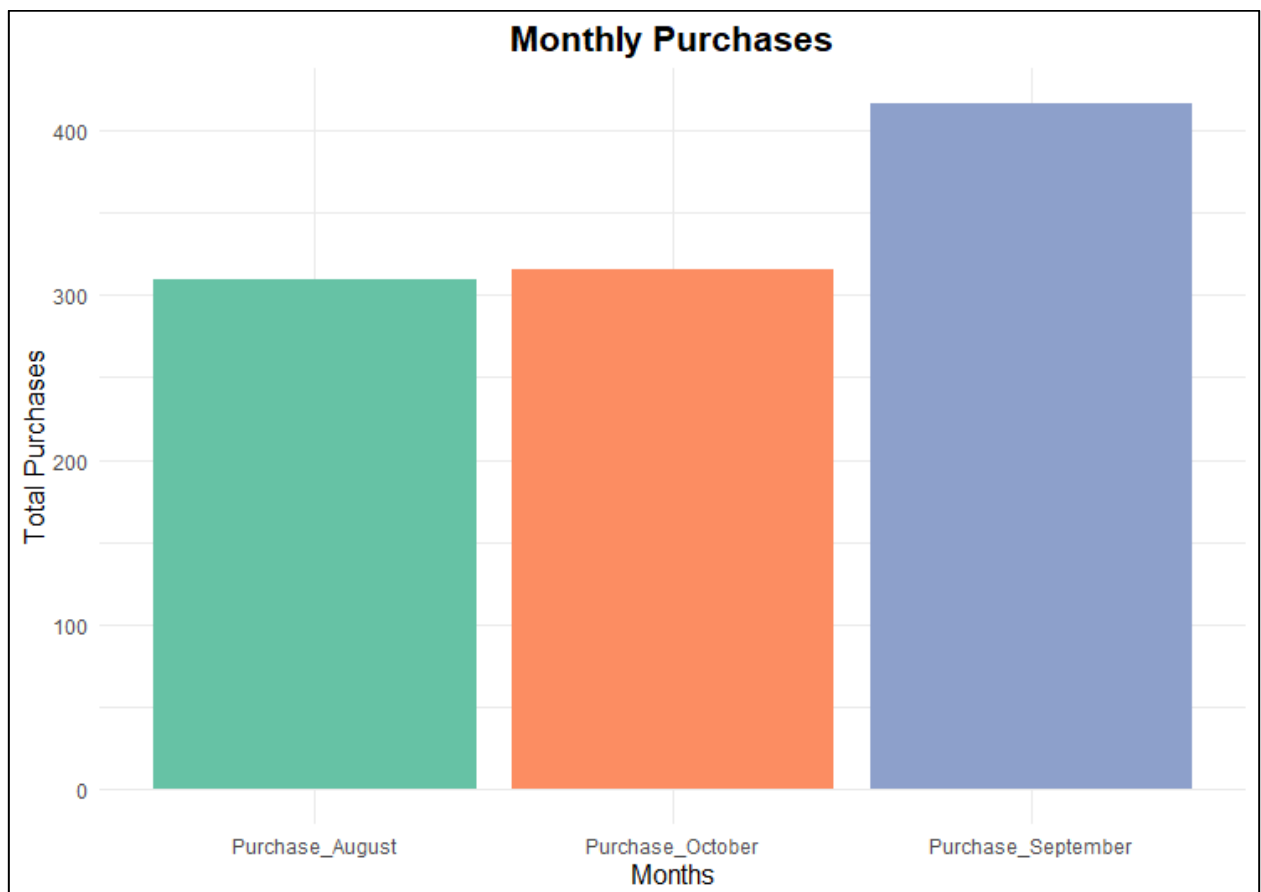
```
ggplot(data, aes(x = Stars)) +  
  geom_histogram(binwidth = 0.1, fill = "blue", color = "black") +  
  labs(title = "Distribution of Product Ratings (Stars)", x = "Stars", y = "Count") +  
  theme_minimal() +  
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```



4.1.2 Monthly Purchases

- **Visualization:** A bar chart displayed the total purchases for August, September, and October.
- **Findings:** September and October showed higher sales compared to August, suggesting increased demand due to seasonal factors like back-to-school shopping.

```
ggplot(monthly_df, aes(x = Month, y = Purchases, fill = Month)) +  
  geom_bar(stat = "identity") +  
  labs(title = "Monthly Purchases", x = "Month", y = "Total Purchases") +  
  theme_minimal() +  
  theme(  
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16),  
    axis.title.x = element_text(size = 12),  
    axis.title.y = element_text(size = 12),  
    legend.position = "none"  
  ) +  
  scale_fill_brewer(palette = "Set2")
```

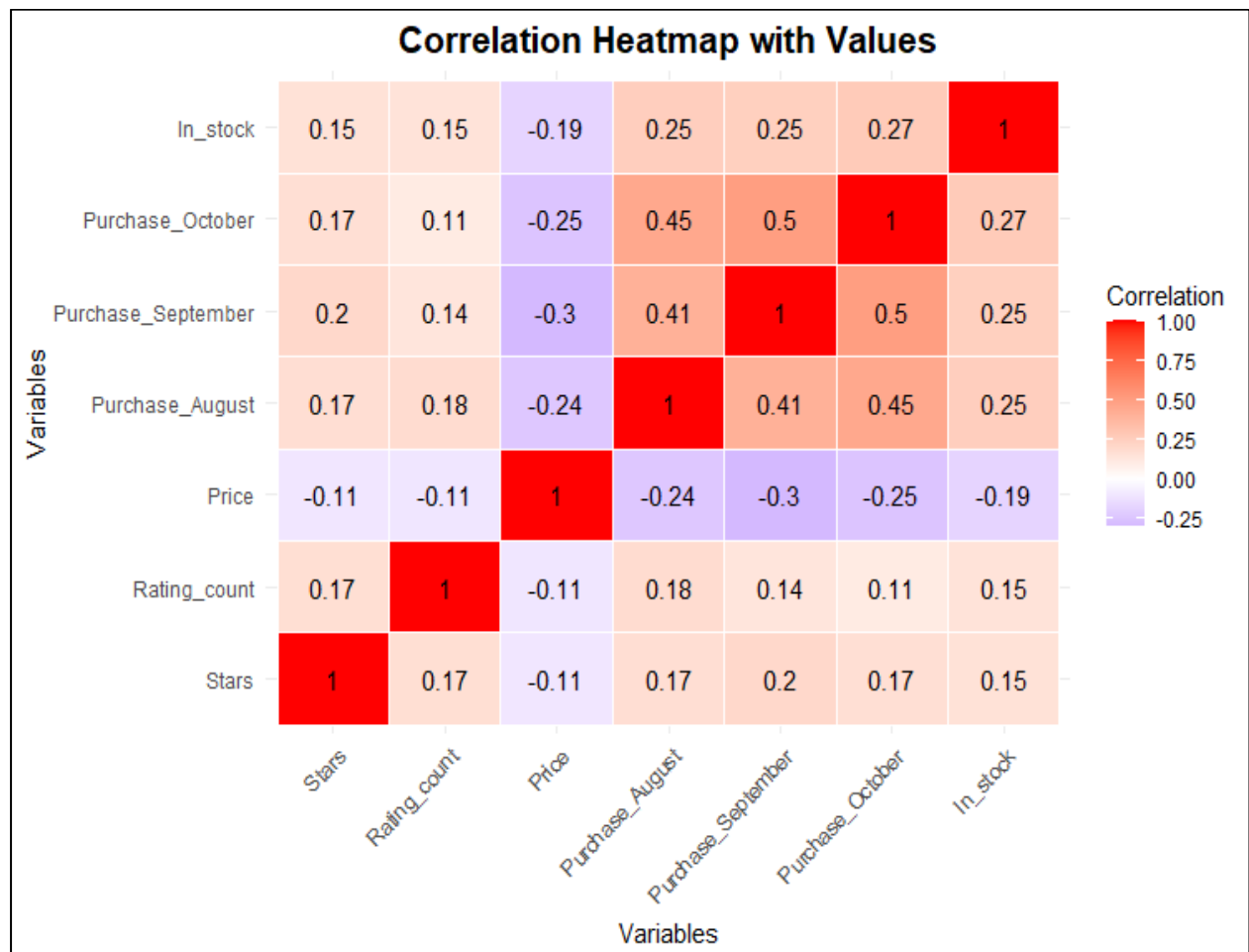


4.1.3 Correlation Analysis

- **Visualization:** A heatmap with numerical correlation values was generated to identify relationships between variables such as Price, Stars, Rating_count, and purchase quantities.

- **Findings:** A negative correlation between Price and purchases indicates price sensitivity among customers. Positive correlations between Rating_count and purchases highlight the importance of customer feedback.

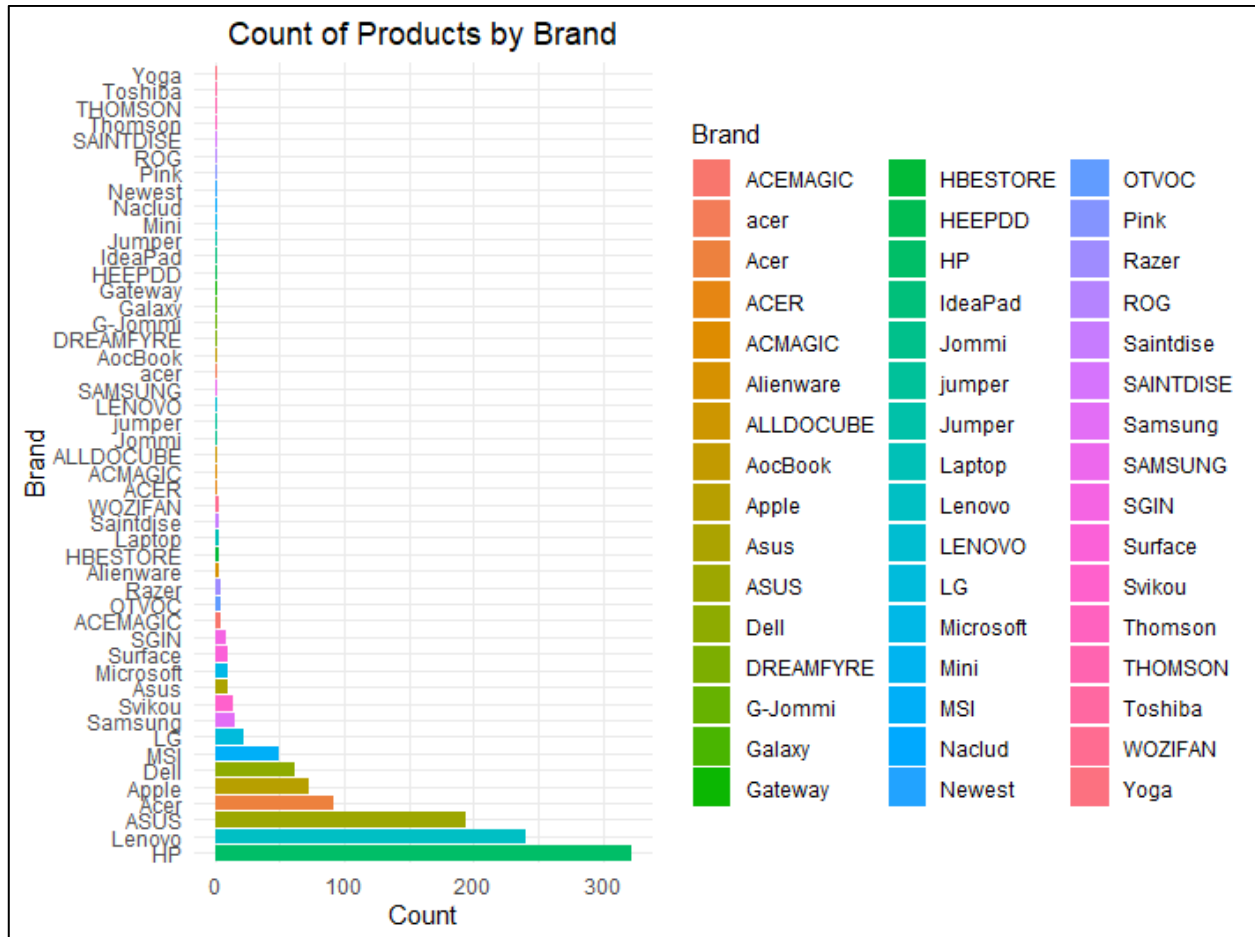
```
ggplot(heatmap, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(value, 2)), color = "black", size = 4) +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white", midpoint = 0) +
  labs(title = "Correlation Heatmap with Values", x = "Variables", y = "Variables") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 16, face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
)
```



4.1.4 Product Count by Brand

- **Visualization:** A horizontal bar chart showed the number of products available for each brand.
- **Findings:** Leading brands like HP and Dell had the highest representation, reinforcing their popularity in the laptop market.

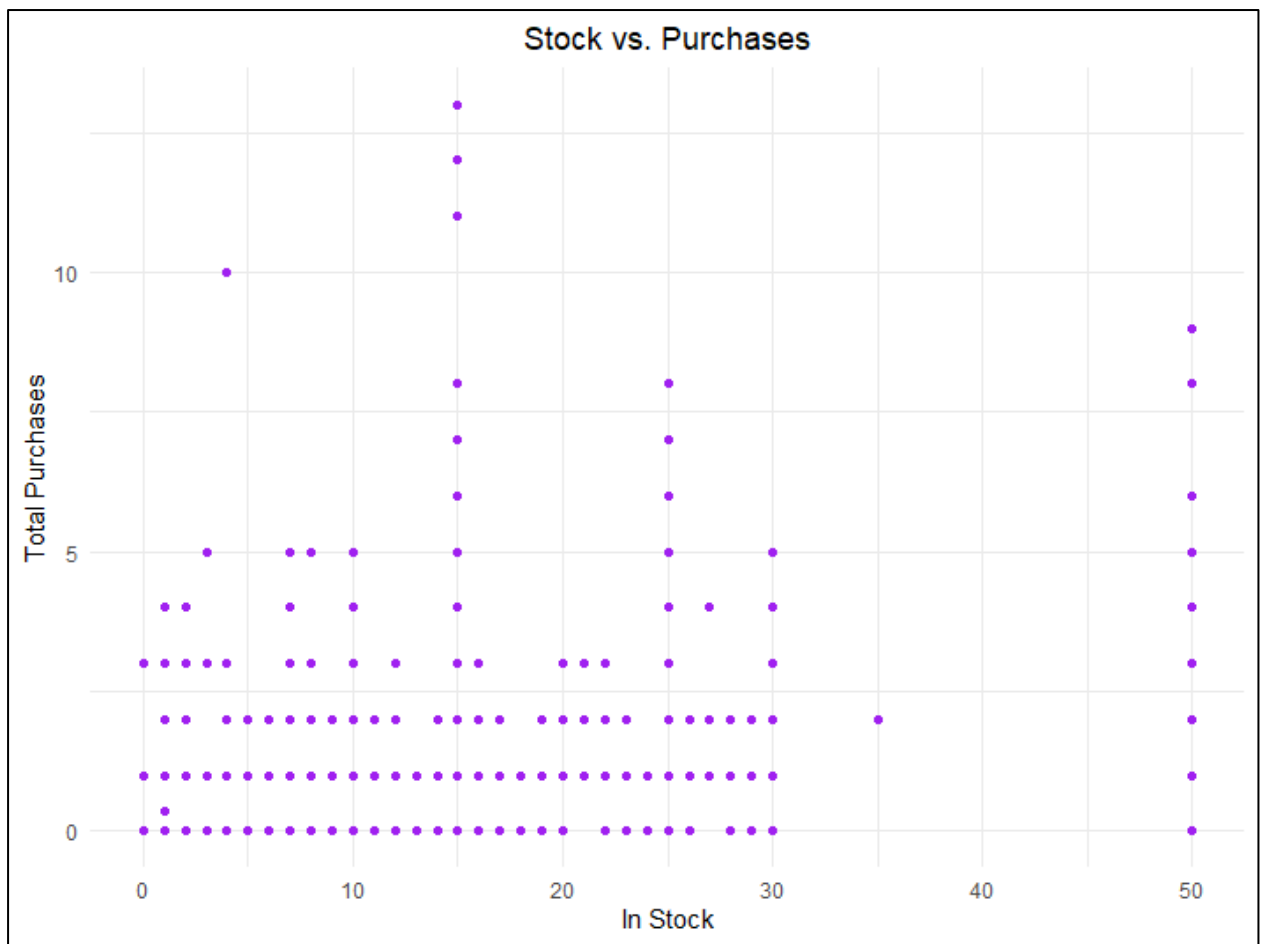
```
ggplot(brand_counts, aes(x = reorder(Brand, -Count), y = Count, fill = Brand)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  labs(title = "Count of Products by Brand", x = "Brand", y = "Count") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
  )
)
```



4.1.5 Stock vs. Purchases

- **Visualization:** A scatter plot compared inventory levels (In_stock) with total purchases (Total_Purchases).
- **Findings:** Products with moderate stock levels were associated with higher purchases, suggesting an optimal inventory threshold for maximizing sales.

```
ggplot(data, aes(x = In_stock, y = Total_Purchases)) +
  geom_point(color = "purple") +
  labs(title = "Stock vs. Purchases", x = "In Stock", y = "Total Purchases") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5)
  )
)
```



5. Optimization Models

5.1 What is Linear Regression?

Linear regression is a statistical modeling technique used to understand the relationship between a dependent variable (outcome) and one or more independent variables (predictors). It estimates the coefficients of the predictors to minimize the difference between observed and predicted values, helping identify key factors influencing the outcome and predict future values.

```
> # 1. Linear Regression Model (Predicting Total Purchases)
> lm_model <- lm(In_stock ~ Price + Rating_count + Stars + Purchase_August + Purchase_September + Purchase_October, data = data)
> summary(lm_model)

Call:
lm(formula = In_stock ~ Price + Rating_count + Stars + Purchase_August +
    Purchase_September + Purchase_October, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-27.894  -6.655  -2.253   4.821  41.878

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   10.3834372   0.7716096   13.457  < 2e-16 ***
Price         -0.0009104   0.0003217   -2.830  0.004730 **
Rating_count    0.0006753   0.0002452    2.754  0.005974 **
Stars          0.4106483   0.1610702    2.549  0.010914 *
Purchase_August 2.0179761   0.5821297    3.467  0.000546 ***
Purchase_September 1.3831950  0.5574392    2.481  0.013228 *
Purchase_October 2.5001821   0.5866626    4.262  2.19e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.906 on 1175 degrees of freedom
Multiple R-squared:  0.125,    Adjusted R-squared:  0.1205
F-statistic: 27.97 on 6 and 1175 DF,  p-value: < 2.2e-16
```

Overview of the Results -

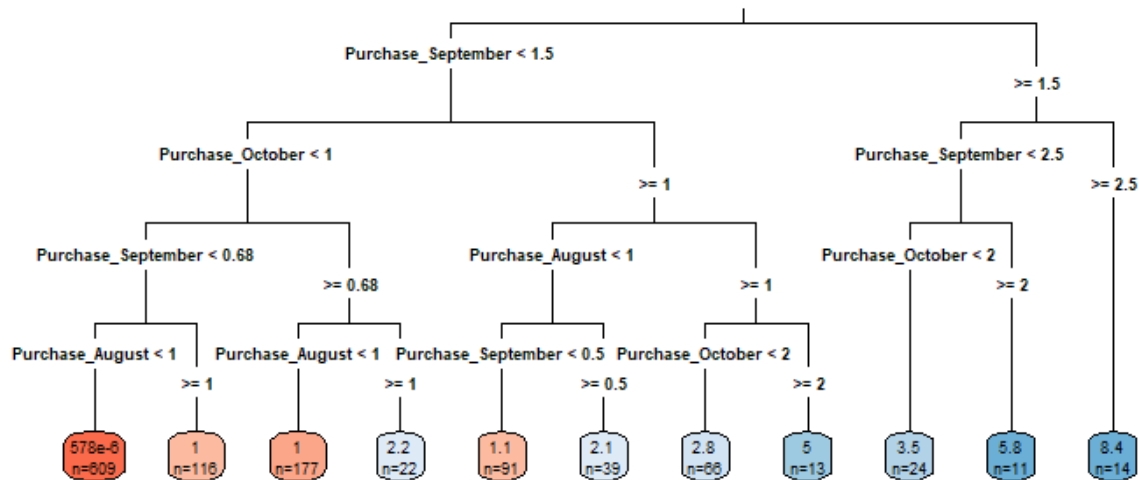
The linear regression model was applied to understand the factors affecting the inventory levels (In_stock) of laptops in the dataset. The predictors included price, ratings, star reviews, and purchase trends over the past three months. Key findings are summarized as follows:

- **Price:** A slight negative relationship was observed between price and inventory levels, indicating that higher-priced laptops tend to have fewer units in stock.
- **Customer Engagement Metrics:** Variables like Rating_count showed significant impacts, with higher ratings in certain groups correlating to lower stock levels, potentially due to higher demand.
- **Purchases Over Months:** The seasonal purchase trends revealed varying impacts on inventory, reflecting how demand patterns influence stock replenishment.

5.2 What is Decision Tree Model?

A decision tree is a machine learning model used for both classification and regression tasks. It splits data into subsets based on feature values, forming a tree-like structure of decisions that lead to an output. Each internal node represents a decision on a feature, each branch represents an outcome, and each leaf node represents a final prediction.

Pruned Decision Tree for Total Purchases



Overview of the Results -

1. Model Fit:

- The tree was trained using rpart in R with the dependent variable Total_Purchases and predictors Price, Rating_count, Stars, In_stock, Purchase_August, Purchase_September, and Purchase_October.
- The dataset contains 1182 observations with no missing values.

2. Tree Complexity:

- The complexity parameter (CP) indicates the cost of adding a new split. As splits are added, the relative error decreases.
- Initial relative error is 1.0, reducing to 0.0845 after 10 splits.

3. Key Predictors:

- Purchase_September:** The most significant variable, contributing 52% importance.
- Purchase_October:** Second most important, with 27%.
- Purchase_August:** Third in importance, with 18%.
- Other variables (Rating_count, In_stock, Price) have minimal contributions.

4. Node Analysis:

- Node 1: Root node, representing the full dataset (1182 observations), splits based on Purchase_September < 1.5, leading to two child nodes.
- Node 2: Represents 1133 observations with a mean Total_Purchases of 0.684.
- Node 3: Represents 49 observations with a mean Total_Purchases of 5.408.

5. Model Performance:

a. MSE (Mean Squared Error):

- i. Root node: 2.076, reducing as splits are added.

b. Primary Splits:

- i. Majority of splits involve Purchase_September, Purchase_October, and Purchase_August.

6. Insights:

- a. **Temporal Variables Dominate:** Variables indicating purchases in specific months (Purchase_September, Purchase_October, Purchase_August) play a significant role in predicting total purchases.
- b. **Low Influence of Price & Rating:** Variables like Price, Rating_count, and Stars contribute minimally to splits.

7. Model Limitations:

- a. The model emphasizes temporal patterns, which might overfit if purchase patterns vary significantly across timeframes.
- b. Variables with low importance might need re-evaluation for potential collinearity or redundancy.

5.3 What is K-Nearest Neighbors (KNN) ?

K-Nearest Neighbors (KNN) is a simple, non-parametric, and instance-based machine learning algorithm used for both classification and regression tasks. KNN works by identifying the k nearest data points (neighbors) to a query point based on a chosen distance metric (e.g., Euclidean distance) and predicting the target value based on these neighbors.

For classification, the output is the most frequent class among the neighbors, while for regression, the output is the average or weighted average of the neighbors' target values. KNN is particularly useful for problems where the relationship between features and target variables is non-linear.

Overview of the Results -

The KNN model in this implementation was used to predict Total_Purchases based on features like Price, Rating_count, Stars, In_stock, and purchase data for the months of August, September, and October. Here's a summary of the results:

1. Data Preparation:

- a. Features were scaled to ensure all variables had equal weight during distance calculations, as KNN is sensitive to the range of feature values.
- b. The dataset was split into 80% training data and 20% testing data to evaluate the model's generalization.

2. Model Predictions:

- a. The KNN model used $k = 5$ (five nearest neighbors) to predict the Total_Purchases for test data.
- b. Predictions were primarily concentrated in lower purchase categories, with most values being 0 (120 occurrences) and 1 (69 occurrences). Higher purchase values were predicted less frequently (2 for 29 samples, 3 for 13 samples, etc.).

3. Output Insights:

- a. The results suggest that most test samples had low predicted purchase counts, aligning with the data's characteristics or potential class imbalance.
- b. The concentration in lower purchase categories highlights the dominance of low-purchase customers in the dataset, which may influence the model's predictions.
- c. The inclusion of features such as Purchase_August, Purchase_September, and Purchase_October reflects temporal trends, possibly contributing to the prediction patterns.

5.4 What is Random Forest Model?

The **Random Forest** is a powerful machine learning algorithm used for both **classification** and **regression** tasks. It works by creating many decision trees during the training process. Each tree makes a prediction, and the final prediction is made by averaging the outputs (in regression tasks) or voting (in classification tasks). This technique helps improve accuracy and reduce the risk of overfitting, making it one of the most popular algorithms for complex problems.

For regression tasks (predicting a continuous value), the Random Forest predicts the target value (like Total_Purchases in your case) by averaging the predictions from all the trees in the forest. This results in more accurate and stable predictions compared to a single decision tree.

Overview of the Results -

1. Model Formula:

- a. The model uses the formula: $\text{Total_Purchases} \sim \text{Price} + \text{Rating_count} + \text{Stars} + \text{In_stock} + \text{Purchase_August} + \text{Purchase_September} + \text{Purchase_October}$. This means the model is predicting the total purchases (Total_Purchases) based on several factors such as price, rating count, stars, whether the item is in stock, and purchase data from previous months.

2. Model Type:

- a. The model is a **regression** type because as we are predicting a continuous number (total purchases).

3. Number of Trees:

- a. The model used **100 decision trees**. The more trees, the better the model can generalize and make accurate predictions, but it also requires more computing power.

4. Variables Tried at Each Split:

- a. During the building of each tree, the model considers **2 variables** at each decision point. This helps reduce the chances of overfitting and makes the model more generalizable.

5. Mean Squared Residuals:

- a. The **mean squared residuals** tell us how far off our predictions are from the actual values. A smaller number means better performance. In our case, the number is quite low, indicating good model performance.

6. Percentage of Variance Explained:

- a. This tells us how much of the variation in the target variable (total purchases) is explained by the model. In our case, the model explains **95.44%** of the variance, meaning it's capturing almost all the important factors that affect total purchases.

7. Performance Evaluation

- a. **Mean Squared Error (MSE):** The MSE of 0.0232346 tells us that the average difference between predicted and actual values is small. This means the model is making accurate predictions.

Understanding Feature Importance

The **importance** of each feature shows how much each variable contributes to the model's predictions. Here's what we find:

- **Purchase Data (August, September, October):** These are the most important features. They indicate how much was purchased in the past months and are directly tied to predicting future purchases.
- **In-stock Status:** This is another important factor. If an item is in stock, it is more likely to be purchased, which explains why it's highly important.
- **Price and Rating Count:** These features also play an important role, although not as much as the purchase-related data.

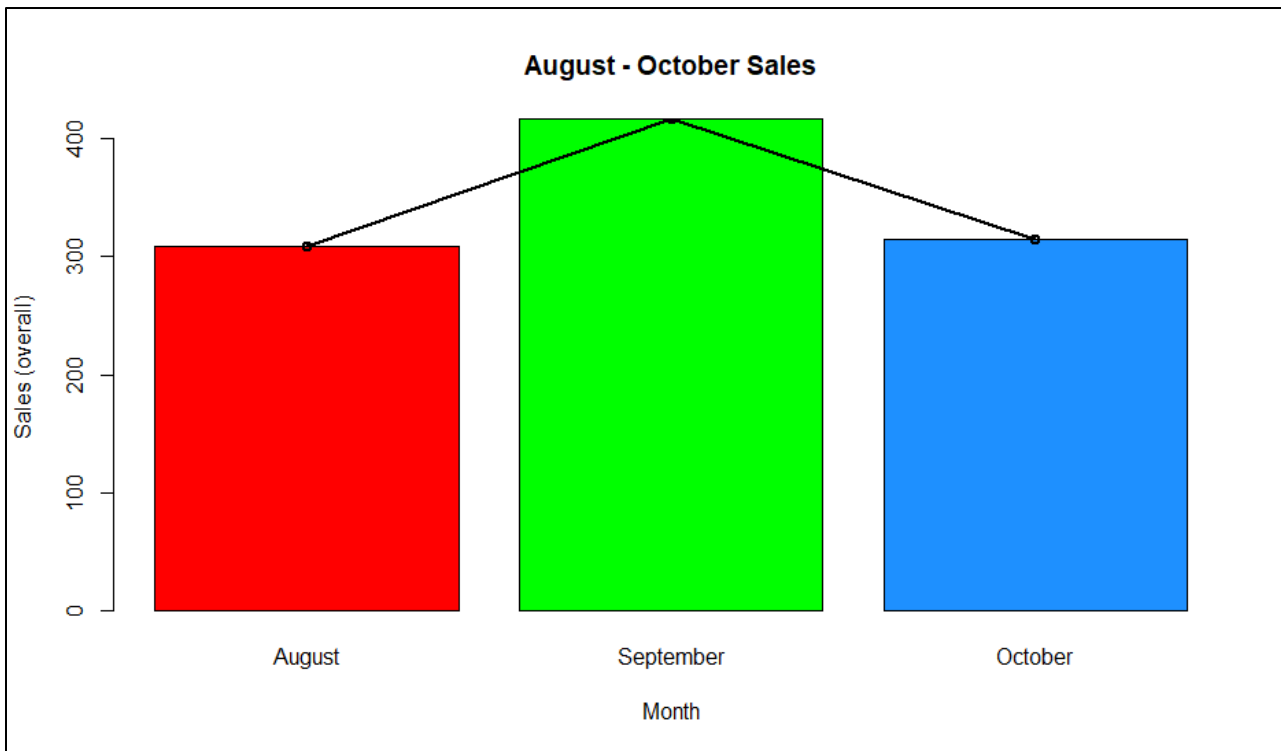
Summary of Key Points

- **MSE:** The Mean Squared Error is **0.0232346**, showing good model accuracy.
- **Variance Explained:** The model explains **95.44%** of the variance in total purchases, indicating it fits the data very well.
- **Important Features:** Features like Purchase_August, Purchase_September, and Purchase_October are the most important for predicting total purchases, which makes sense since they directly relate to recent buying patterns.

5.5 What is Time series Model?

Demand forecasting is the process of predicting future sales using historical data. Time series analysis, a statistical method, is commonly used to forecast future trends based on past observations. By analyzing the patterns in the past sales data for a product or service, we can identify trends, seasonal variations, and cyclic behavior that will help predict future demand.

In our project, we **combined monthly sales data** from August, September, and October for forecasting future demand. Time series forecasting helps in understanding the underlying patterns in the sales data and allows businesses to make informed decisions about inventory, supply chain, and resource planning.



Overview of the Results -

The **Time Series Object** was created by combining the sales data for **August, September, and October** into a single time series object (`combine_month_sales`). We used the `ts()` function to create a time series object with the following properties:

- **Start period:** August 2024
- **Frequency:** 12 (indicating yearly frequency, but since we only have data for three months, it focuses on those months for forecasting).

Next, we visualized the monthly sales data using a **bar plot**, with each month's sales (from August to October) represented by bars, and a line overlaid to show the trend.

Key Insights from the Visualization:

- **Sales Trends:** The chart helps visualize the sales trend for the three months. By observing the heights of the bars and the line connecting them, we can determine if there is any noticeable growth, decline, or consistency in sales.
- **Seasonality or Fluctuation:** While we only have data for three months, observing these trends can help us understand the seasonal effect or fluctuations in sales.

5.6 What is ABC Classification?

ABC Classification is a method used in inventory management to categorize items based on their importance, with "A" items being the most valuable, "B" items being of medium importance, and "C" items being the least valuable. This technique is widely used to prioritize inventory management and focus resources on the most important items. The ABC classification is based on the Pareto principle, which suggests that a small percentage of items (or customers) often contribute to a large percentage of the total sales or value.

The steps for implementing ABC classification include:

1. **Sorting** the items based on their total sales (or total purchases in this case), from highest to lowest.
2. **Calculating the cumulative sales** for each item, relative to the total sales.
3. **Categorizing** items into three classes based on the cumulative sales:
 - a. **Class A:** The top 10% of items (the most valuable in terms of purchases).
 - b. **Class B:** The next 30% of items (medium importance).
 - c. **Class C:** The remaining 60% (the least important).

Overview of the Results -

For this analysis, we performed the ABC classification based on the total purchases of products. The total purchases were sorted in descending order, and the cumulative sales were calculated. The items were then classified into three categories (A, B, and C) based on the cumulative sales.

Results of ABC Classification:

- **Class A:** 10 items, which account for the top 10% of the total purchases.
- **Class B:** 77 items, representing the next 30% of the total purchases.
- **Class C:** 1,095 items, accounting for the remaining 60% of the purchases.

This classification allows us to focus on the **Class A** items, which are the most valuable in terms of sales, while managing the **Class B** and **Class C** items with less intensity.

5.7 What is Safety Stock Calculation?

Safety stock is an additional quantity of inventory kept to mitigate the risk of stockouts caused by uncertainties in demand and lead time. It is used to account for variability in demand and supply during the time it takes to replenish stock (lead time). The goal is to ensure that inventory levels are sufficient to meet customer demand even during unexpected fluctuations.

The formula for calculating safety stock is:

$$\text{Safety Stock} = Z \times \sigma_{\text{demand}} \times \sqrt{L}$$

Where:

- **Z** is the Z-factor, representing the desired service level (for a 95% service level, $Z = 1.65$).
- σ_{demand} is the standard deviation of demand (the variability in total purchases).
- **L** is the lead time (in months, weeks, or days, depending on the context).

In this case, the Z-factor for a 95% service level is typically set to 1.65, which corresponds to a confidence level where there is a 95% chance of not experiencing a stockout.

Overview of the Results-

In this analysis, we calculated the **safety stock** for the product based on the total purchases data. Here are the steps and results:

- **Average Demand:** The mean of the total purchases was calculated as a baseline for the expected demand.

- **Standard Deviation of Demand:** The variability in total purchases (standard deviation) was calculated, which reflects how much demand fluctuates.
- **Lead Time:** A lead time of 2 months was assumed, which is the time it takes to replenish stock.
- **Z-Factor:** A Z-factor of 1.65 was used, which corresponds to a 95% service level (i.e., there is a 95% chance that stock will be available during the lead time).

Based on these values, the **safety stock** was calculated as 3.36 units.

- The calculated **safety stock** of 3.36 units means that, in addition to the regular stock, it is recommended to keep an extra 3.36 units of inventory to handle variability in demand during the 2-month lead time.
- This safety stock helps ensure that you can meet customer demand even during periods of unexpectedly high purchases or delays in replenishment.
- This result is useful for **ensuring product availability** and avoiding stockouts, especially for high-demand products where customer expectations are high, and the consequences of stockouts could be significant.

5.8 What is Reorder Point Calculation?

Reorder point (ROP) is the inventory level at which a new order should be placed to replenish stock before running out. It ensures that inventory is reordered in time to avoid stockouts, considering both the demand during lead time and the safety stock. The reorder point considers the average demand during the lead time and the safety stock that accounts for variability in demand.

The formula for calculating the reorder point is:

$$\text{Reorder Point} = (\text{Average Demand per Unit Time} \times \text{Lead Time}) + \text{Safety Stock}$$

Where:

- **Average Demand per Unit Time** is the average number of units sold during the lead time.
- **Lead Time** is the time it takes to receive the next order after placing it (in this case, 2 months).
- **Safety Stock** is the buffer stock kept covering demand variability during the lead time.

The reorder point ensures that the stock is reordered at the right time to avoid stockouts and maintain product availability.

Overview of the Results -

In this analysis, the **reorder point** was calculated for the product based on the following parameters:

- **Average Demand:** The average total purchase across months was calculated earlier.
- **Lead Time:** The lead time for replenishment is set to 2 months.
- **Safety Stock:** The safety stock calculated in the previous step was 3.36 units.

Using the formula for reorder point:

$$\text{Reorder Point} = \text{Average Demand} \times \text{Lead Time} + \text{Safety Stock}$$

The **reorder point** was calculated as **5.12 units**.

- The **reorder point** of 5.12 units means that when the inventory reaches around 5.12 units, a new order should be placed to replenish stock, considering the demand during the lead time and the safety stock.
- This calculation ensures that the inventory levels are maintained at optimal levels, reducing the risk of stockouts while avoiding excessive inventory that could lead to higher holding costs.

6. Recommendation

Effective inventory management is essential for reducing costs, meeting customer demand, and improving operations. This section highlights key strategies to optimize inventory and enhance supply chain efficiency.

6.1. Accurate Demand Forecasting

- **Importance:** Predicting customer demand accurately helps in planning inventory levels, preventing overstocking or stockouts.
- **Why:** Overestimating demand leads to excess inventory and higher holding costs, while underestimating results in lost sales and dissatisfied customers. Advanced forecasting techniques, such as ARIMA and machine learning, ensure better predictions.

6.2. Inventory Classification (ABC Analysis)

- **Importance:** Prioritizing inventory management efforts based on product value and sales frequency.
- **Why:** By categorizing items into A (high-value, low quantity), B (moderate-value), and C (low-value, high quantity), businesses can focus resources on high-impact products, improving operational efficiency and cost control.

6.3. Safety Stock Calculation

- **Importance:** Maintaining a buffer stock prevents stockouts during unexpected demand surges or supply chain delays.
- **Why:** It ensures continuous operations, minimizes revenue loss, and enhances customer satisfaction, especially for critical or high-demand items.

6.4. Reorder Point Optimization

- **Importance:** Determining the right time to replenish inventory based on lead times and demand variability.
- **Why:** Optimized reorder points ensure timely restocking, reducing the risk of running out of stock or carrying excessive inventory.

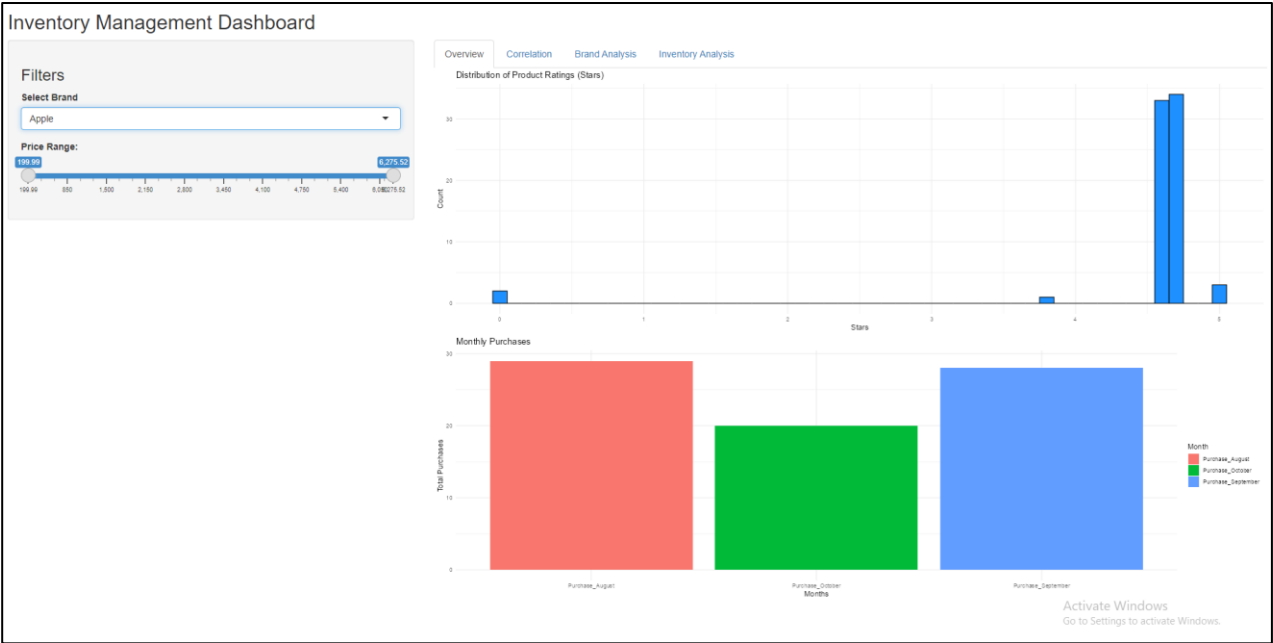
6.5. Continuous Monitoring and Analytics

- **Importance:** Regularly tracking inventory levels, sales trends, and supplier performance helps in proactive decision-making.
- **Why:** Continuous monitoring ensures adaptability to market changes, prevents outdated inventory, and improves supply chain responsiveness.

6.6. Creating a Dashboard

- **Importance:** A dashboard is crucial for any inventory management system as it serves as an interactive, visual interface that transforms complex data into easily digestible insights. They help to understand important patterns, such as stock imbalances or seasonal demand fluctuations, at a glance, making it easier to respond proactively to business needs.
- **Why:** This type of dashboard is important because it centralizes essential data into a single interface, simplifying the monitoring of inventory and sales performance. Brand analysis helps identify top-performing products, while inventory analysis ensures that stock levels

align with purchasing trends, preventing stockouts or overstocking.



7. Conclusion

In conclusion, the models and techniques applied in this project provide valuable insights for optimizing inventory management. The Random Forest model, for instance, demonstrated high predictive accuracy, explaining a substantial portion of the variance in total purchases. The K-Nearest Neighbors (KNN) model also contributed by providing useful predictions, though with slightly higher error metrics compared to Random Forest. Demand forecasting using ARIMA allowed for effective time-series analysis, helping predict future purchase patterns, particularly for key months. The integration of these models supports data-driven decision-making, providing a solid foundation for improving inventory levels, minimizing stockouts, and reducing excess inventory. Overall, the combination of machine learning models and forecasting techniques offers a comprehensive approach to optimizing inventory management, leading to better resource allocation, cost control, and improved customer satisfaction.

8. References

- Amazon Canada dataset (Brantford location). Retrieved from [[Amazon.ca: 4 Stars & Up - New / Laptop Computers / Computers & Accessories: Electronics](#)].
- R Core Team (2021). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. URL: <https://www.R-project.org>
- **ggplot2** - Wickham, H. (2016). ggplot2: Elegant Graphics for Data Analysis. Springer. DOI: 10.1007/978-3-319-24277-4
- **dplyr** - Wickham, H., Francois, R., Henry, L., & Müller, K. (2021). dplyr: A Grammar of Data Manipulation. R package version 1.0.7. Retrieved from <https://cran.r-project.org/web/packages/dplyr/index.html>
- **randomForest** - Liaw, A., & Wiener, M. (2002). Classification and Regression by randomForest. R news, 2(3), 18-22. Retrieved from <https://cran.r-project.org/web/packages/randomForest/randomForest.pdf>
- **forecast** - Hyndman, R. J., & Khandakar, Y. (2008). Automatic Time Series Forecasting: The forecast Package for R. Journal of Statistical Software, 27(3), 1-22. Retrieved from <https://www.jstatsoft.org/v27/i03>
- **rpart** - Therneau, T. M., & Atkinson, E. (2021). rpart: Recursive Partitioning and Regression Trees. R package version 4.1-15. Retrieved from <https://cran.r-project.org/web/packages/rpart/rpart.pdf>